

CHAPTER 18

PRACTICAL

ELECTRICITY

4.4 Practical electricity

4.4.1 Uses of electricity

- 1 State common uses of electricity, including heating, lighting, battery charging and powering motors and electronic systems
- 2 State the advantages of connecting lamps in parallel in a lighting circuit
- 3 Recall and use the equation
power = current $\times$ voltage
 $P = IV$
- 4 Recall and use the equation
energy = current $\times$ voltage $\times$ time
 $E = IVt$
- 5 Define the kilowatt-hour (kWh) and calculate the cost of using electrical appliances where the energy unit is the kWh

4.4.2 Electrical safety

- 1 State the hazards of:
 - (a) damaged insulation
 - (b) overheating cables
 - (c) damp conditions
 - (d) excess current from overloading of plugs, extension leads, single and multiple sockets when using a mains supply
- 2 Explain the use and operation of trip switches and fuses and choose appropriate fuse ratings and trip switch settings
- 3 Explain what happens when a live wire touches a metal case that is earthed
- 4 Explain why the outer casing of an electrical appliance must be either non-conducting (double-insulated) or earthed
- 5 Know that a mains circuit consists of a live wire (line wire), a neutral wire and an earth wire and explain why a switch must be connected into the live wire for the circuit to be switched off safely
- 6 Explain why fuses and circuit breakers are connected into the live wire

1. O/N/23/11/Q28

Three students make comments about appliances with double insulation.

student P Appliances with double insulation do not need an earth wire.

student Q Double insulation protects the user of the appliance from an electric shock.

student R The circuits of appliances with double insulation do not contain fuses.

Which students are correct?

- A** P, Q and R **B** P and Q only **C** P and R only **D** Q and R only

2. M/J/23/12/Q31

The power consumption of a microwave oven is 1500 W.

The cost of 1.0 kWh of electrical energy is 24 cents.

What is the cost of using the microwave oven for 10 minutes?

- A** 6 cents
B 36 cents
C 60 cents
D 360 cents

3. M/J/23/11/Q31

Three students each suggest a consequence of wiring lamps in parallel rather than in series.

student 1 If one lamp breaks, the other lamps stay lit.

student 2 The lamps can be switched on and off separately.

student 3 The p.d. across each lamp is equal to the p.d. across the supply.

Which students are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

4. O/N/22/11/Q32

Why is a fuse used in an electrical appliance?

- A** to earth the appliance
B to protect the appliance
C to change the power of the appliance
D to change the voltage supplied to the appliance

5. O/N/22/12/Q33

Which device is used to convert electrical energy to kinetic energy?

- A iron
- B kettle
- C lamp
- D motor

6. M/J/22/12/Q35

When the flash on a camera is used, a charge of 1.5 C flows for 0.0030 s through the flash lamp.

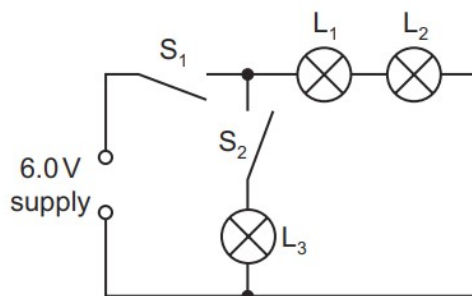
The average voltage across the flash lamp is 3600 V.

What is the electrical energy supplied to the flash lamp and what is the average power supplied?

	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400	1.8×10^6

7. M/J/22/12/Q36

The diagram shows an electric circuit.

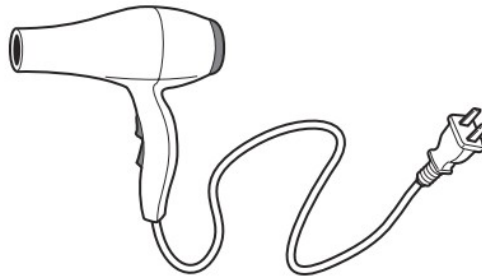


Which lamps light when switch S_1 is closed and S_2 is open?

- A L_1 only
- B L_3 only
- C L_1 and L_2 only
- D L_1 , L_2 and L_3

8. M/J/22/12/Q37

The diagram shows a hairdryer.



Which statement explains why the hairdryer does **not** need an earth wire?

- A** It has a live wire.
- B** It has a neutral wire.
- C** It has double insulation.
- D** The cable is insulated.

9. M/J/22/11/Q35

A metal kettle is plugged into a mains socket. The plug contains a suitable fuse.

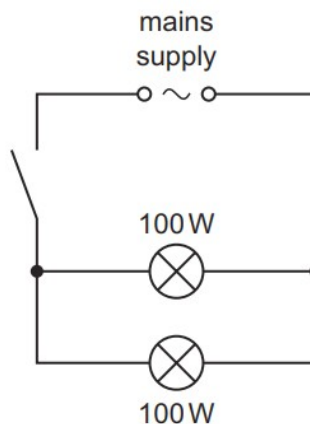
The kettle's cable is damaged and the fuse blows.

Why does the fuse blow?

- A** The bare earth wire in the cable touches the bare live wire.
- B** The metal casing of the kettle touches the bare earth wire.
- C** The bare neutral wire in the cable touches the bare earth wire.
- D** The live wire breaks without touching any other wire.

10. O/N/21/11/Q32

Two 100 W lamps are connected in parallel to the mains supply, as shown.



How much electrical energy is supplied by the mains when the switch is closed for 36 minutes?

- A** 0.030 kWh **B** 0.060 kWh **C** 0.12 kWh **D** 7.2 kWh

11. O/N/21/11/Q33

The cost of electricity is 9.0 cents/kWh.

It takes 0.060 hours and costs 1.2 cents to boil the water in a kettle.

What is the power rating of the kettle?

- A** 0.45 kW **B** 0.65 kW **C** 2.2 kW **D** 130 kW

12. M/J/21/12/Q35

A 100 W lamp is switched on for five hours each day for three weeks.

The cost of one unit of electricity is \$0.24.

How much does it cost to run the lamp for this time?

- A** \$0.36 **B** \$0.84 **C** \$2.52 **D** \$25.20

13. M/J/21/12/Q36

A three-pin mains plug contains a fuse. A device with double insulation has a cable connected to the plug.

Which part of the plug is **not** connected to the cable?

- A** earth pin
B fuse
C live pin
D neutral pin

14. M/J/21/11/Q31

The information on the back of an electric room heater is shown.

rating 220–240 V
~50 Hz
4.2 A

What is a suitable fuse rating for this room heater?

- A** 4.0 A **B** 4.2 A **C** 5.0 A **D** 13.0 A

15. O/N/20/11/Q39

A student calculates the amount of energy used by an electric heater.

What is the equation for calculating the energy E in kWh?

- A** $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (second)}$
- B** $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (hour)}$
- C** $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (second)} \div 1000$
- D** $E = I \text{ (ampere)} \times V \text{ (volt)} \times t \text{ (hour)} \div 1000$

16. O/N/20/12/Q33

A 240 V mains circuit contains eight 60 W lamps in parallel. At the time when the lamps are switched on, the filaments are cold and the current is four times as large as the final steady current in the circuit.

What is the initial current supplied by the mains?

- A** 0.25 A
- B** 1.0 A
- C** 2.0 A
- D** 8.0 A

17. M/J/20/11/Q40

A kettle is connected to the 240 V mains supply using a plug containing a 13 A fuse.

The kettle contains water.

When it is switched on, the fuse blows. This happens again after a new fuse is fitted.

Someone replaces the fuse with a nail, and the kettle works.

What else might happen as a result of replacing the fuse with a nail?

- A** A very large current overloads the wiring, causing a fire.
- B** The kettle boils the water less quickly.
- C** The kettle uses more energy to boil the water.
- D** The water boils at a higher temperature.

18. O/N/19/11/Q36

A lamp is connected to a mains plug.

Where are the switch and the fuse connected so that the lamp is safe to use?

	switch	fuse
A	live wire	live wire
B	live wire	neutral wire
C	neutral wire	live wire
D	neutral wire	neutral wire

19. O/N/19/12/Q36

A mains electric circuit is fitted with a circuit breaker rather than a fuse.

Where is the circuit breaker connected and what happens when the current is too large?

	A circuit breaker is connected in...	When the current is too large...
A	the live wire.	a thin wire melts and breaks the circuit.
B	the live wire.	an electromagnet opens a switch.
C	the neutral wire.	a thin wire melts and breaks the circuit.
D	the neutral wire.	an electromagnet opens a switch.

20. M/J/19/12/Q35

The cable to an electric cooker contains a live wire, a neutral wire and an earth wire.

When the cooker is working correctly, in which wires are the currents equal?

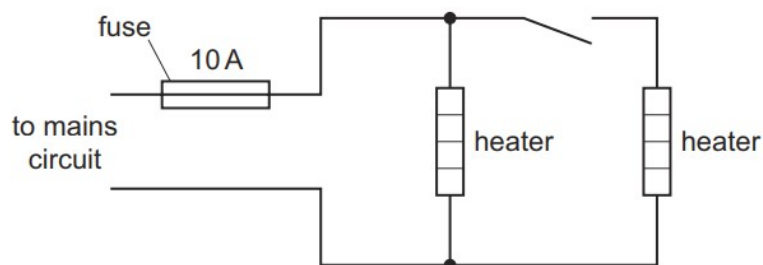
- A** the live, the neutral and the earth
- B** the live and the earth only
- C** the neutral and the earth only
- D** the neutral and the live only

21. M/J/19/12/Q36

The manufacturer of an electrical heater states that only one heater must be connected to the mains wiring through any fuse.

The current in the circuit with only one heater is 9 A. The fuse rating is 10 A.

Despite the warning, a second identical heater is added, as shown. When the switch is closed the fuse blows.



The fuse is replaced with a 20 A fuse.

What happens when the switch is closed?

- A** Each heater only receives half the correct current.
- B** Each heater receives twice the correct current.
- C** The circuit wiring connecting the heaters to the mains may overheat.
- D** The fuse melts when the heaters are switched on.

22. O/N/18/11/Q31

The power of an electrical heater is 3.0 kW.

The heater is switched on for a time of 2 hours and 30 minutes.

The cost of using the heater for this time is 120 c.

What is the cost of 1.0 kWh of electrical energy?

- A** 16 c **B** 17 c **C** 40 c **D** 48 c

23. O/N/18/11/Q32

In a lighting circuit, the switch is placed in the live wire.

Why is this?

- A** A lamp in the circuit can be isolated from the supply.
B The fuse is in the neutral wire.
C The switch does not work in the neutral wire.
D Too much current flows in the earth wire.

24. M/J/18/11/Q31

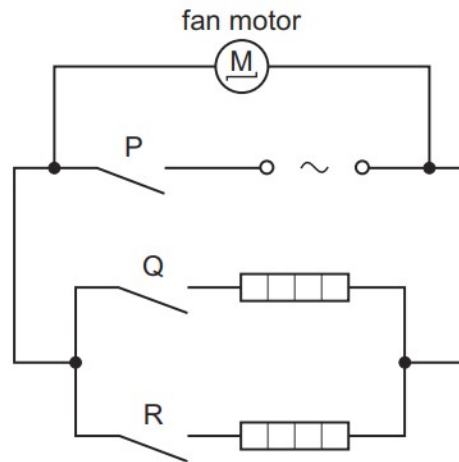
An electrical appliance is plugged into a socket in the wall. The plug contains a fuse.

What is the main purpose of the fuse?

- A** to earth the appliance
B to earth the plug
C to protect the user from electric shock
D to protect the wiring from overheating

25. O/N/17/11/Q32

The diagram shows the circuit for a hair-dryer.



The fan motor has a power rating of 0.10 kW and the heaters each have a rating of 0.40 kW. The cost of electricity is 8.0 cents/kWh.

What is the cost of running the hair-dryer for two hours with switches P and Q closed and switch R open?

- A** 1.6 cents **B** 4.0 cents **C** 6.4 cents **D** 8.0 cents

26. O/N/17/11/Q33

Why are the metal casings of electrical appliances earthed?

- A** to complete the circuit
- B** to ensure the casing is not at a dangerous voltage
- C** to ensure the fuse blows when the current in the appliance is too large
- D** to protect the appliance from overheating

27. O/N/16/12/Q35

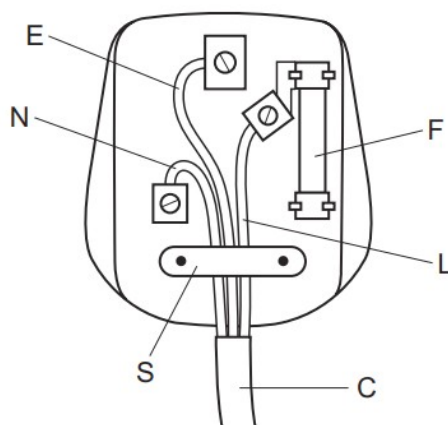
The headlamp circuit in a car includes a 7A fuse.

What is the purpose of the fuse?

- A** It increases the resistance of the circuit.
- B** It makes it easier for the current to flow.
- C** It protects the wiring from overheating.
- D** It reduces the voltage across the lamp.

28. M/J/16/12/Q30

The diagram shows the wiring of a three-pin mains plug. There is an error in the diagram.



What is the error?

- A** The cable cover C is not under the clip S.
- B** The earth wire E is connected to the wrong terminal.
- C** The fuse F is connected to the live wire L.
- D** The live wire L is connected to the wrong end of the fuse F.

29. M/J/16/12/Q31

A 100 W lamp is switched on for five hours each day for three weeks.

The cost of one unit of electricity is \$0.24.

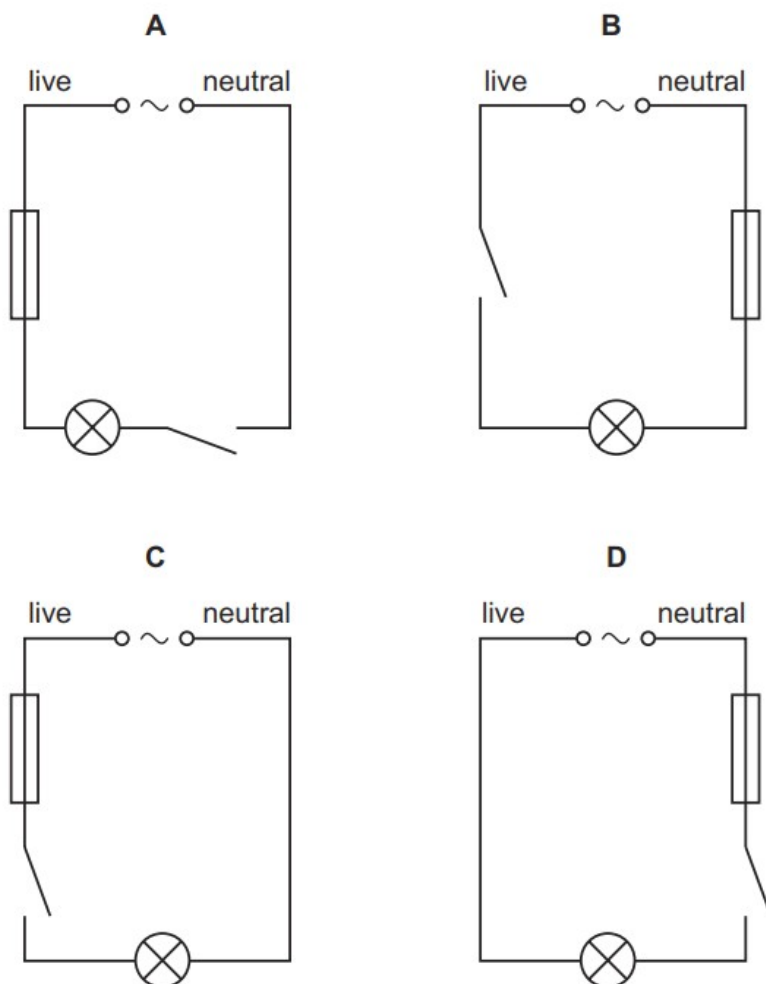
How much does it cost to run the lamp for this time?

- | | | | |
|-----------------|-----------------|-----------------|------------------|
| A \$0.36 | B \$0.84 | C \$2.52 | D \$25.20 |
|-----------------|-----------------|-----------------|------------------|

30. M/J/16/11/Q33

A lamp is connected to the a.c. mains supply in series with a switch and a fuse.

Which circuit shows these components wired correctly?



31. O/N/15/12/Q34

An immersion heater is labelled 24 V, 120 W.

What is the current in the heater when it is connected to a 24 V supply?

- A** 0.20 A **B** 5.0 A **C** 24 A **D** 120 A

32. M/J/15/11/Q31

When the flash on a camera is used, a charge of 1.5 C flows through the flash-tube in 0.0030 s.

The average voltage across the flash-tube is 3600 V.

What is the electrical energy supplied to the flash-tube and what is the average power supplied?

	energy / J	power / W
A	2400	7.2
B	2400	800 000
C	5400	16.2
D	5400	1.8×10^6

33. O/N/14/12/Q28

There is a current of 0.25 A in a lamp connected to a 240 V supply.

What is the input power to the lamp?

- A** 15 W **B** 60 W **C** 240 W **D** 960 W

34. O/N/14/12/Q29

Which unit measures the energy input to an electrical appliance?

- A** ampere
B kilowatt-hour
C volt
D watt

35. O/N/14/12/Q30

An electric lamp is marked 0.5 A. It is connected to a socket marked '30 A maximum'.

Which fuse is best to use in the lamp?

- A** 0.5 A **B** 3 A **C** 30 A **D** 40 A

36. M/J/14/12/Q33

An appliance uses a current of 3 A.

Which row is correct for the fuse in this appliance?

	most suitable fuse rating / A	fuse connected in
A	5	earth wire
B	5	live wire
C	13	earth wire
D	13	live wire

37. M/J/14/11/Q32

One kilowatt-hour of energy costs \$0.24.

How much does it cost to run a 2 kW heater for three hours?

- A** \$0.24 **B** \$0.48 **C** \$0.72 **D** \$1.44

38. M/J/13/12/Q36

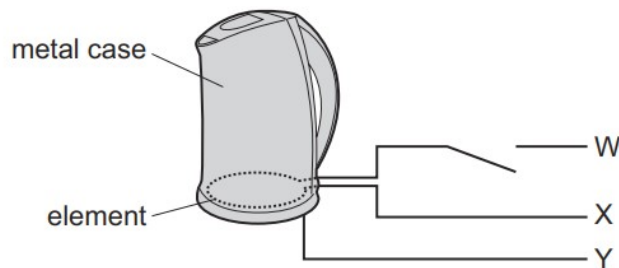
Five electrical appliances are connected to the same socket and there is a very large current.

Why is this dangerous?

- A** The fuses blow in the appliances.
B There is a greater risk of an electrical shock.
C There is overheating in each appliance.
D There is overheating in the socket.

39. M/J/13/11/Q33

The diagram represents part of a household circuit containing an electric kettle.



Which row correctly identifies the wires W, X and Y?

	W	X	Y
A	earth	live	neutral
B	live	neutral	earth
C	live	earth	neutral
D	neutral	live	earth

40. O/N/12/11/Q31

A lamp connected to a 12 V supply converts energy at a rate of 36 W.

How much energy will be converted in 10 s?

- A** 30 J **B** 36 J **C** 120 J **D** 360 J

41. O/N/12/11/Q32

An immersion heater is labelled 12 V, 60 W.

What is the current in the heater when connected to a 12 V supply?

- A** 0.20 A **B** 5.0 A **C** 12 A **D** 60 A

42. O/N/12/12/Q31

A lamp is rated at 60 W on a 240 V supply.

What is the current in the lamp when used normally?

- A** 0.25 A **B** 4.0 A **C** 60 A **D** 180 A

43. O/N/12/12/Q32

The cost of electricity is 16 c per kWh.

What is the cost of using a 3 kW heater for 4 hours?

- A** $\frac{3 \times 4}{16}$ c **B** $\frac{3 \times 16}{4}$ c **C** $4 \times 3 \times 16$ c **D** $4 \times 3 \times 60 \times 16$ c

44. M/J/12/12/Q31

A lamp is rated at 12 V, 600 mW.

What is the current in the lamp?

- A** 20 mA **B** 50 mA **C** 2.0 A **D** 5.0 A

45. M/J/12/12/Q32

A water heater uses 6 kW of electric power when connected to a 240 V circuit.

Which fuse is **most** suitable for use in this circuit?

- A** 5 A **B** 13 A **C** 30 A **D** 50 A

46. M/J/12/11/Q31

A lamp connected to a 12 V supply converts energy at a rate of 36 W.

How much energy will be converted in 10 s?

- A** 30 J **B** 36 J **C** 120 J **D** 360 J

47. M/J/12/11/Q32

What causes the fuse to blow in a mains electrical circuit?

- A** a person touches the live wire
B a person touches the neutral wire
C the live wire touches the earth wire
D the neutral wire touches the earth wire

48. O/N/11/11/Q30

The current in a filament lamp is 0.25 A when working normally. The lamp is connected to a plug and the mains a.c. supply.

When the lamp is switched on, it does not light.

What is a possible cause for this?

- A** The earth wire in the plug is not connected.
- B** The fuse in the plug is 3 A .
- C** The lamp only works on a d.c. power supply.
- D** The live wire in the plug is not connected.

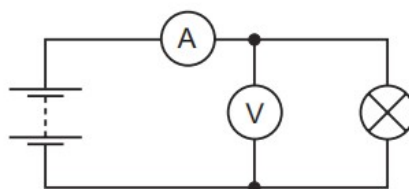
49. O/N/11/11/Q31

Which appliance used on a 240 V supply is **most** likely to burn out its fuse?

	appliance	fuse
A	100 W lamp	1 A
B	1 kW vacuum cleaner	5 A
C	2 kW fan heater	3 A
D	3 kW electric fire	13 A

50. M/J/11/12/Q32

The diagram shows a circuit.



The lamp is a 12 W lamp and is working at normal brightness.

What are the readings on the meters?

	voltmeter reading / V	ammeter reading / A
A	6	0.5
B	12	0.5
C	12	1
D	24	2

51. O/N/10/11/Q31

Many electrical appliances have metal cases.

To prevent the case from becoming 'live', with the possibility of an electric shock, the earth wire of the electric cable is attached to the case.

How does the earth wire prevent an electric shock?

- A** It allows a current to flow to earth, so that the appliance continues working.
- B** It allows a large current to flow to earth, blowing the fuse.
- C** It prevents the fuse from blowing.
- D** It reduces the current to a safe level.

52. M/J/10/12/Q29

The metal case of an electric heater is earthed. The plug to the heater contains a 5 A fuse. There is a current of 4 A when the heater works normally.

The cable to the heater becomes so worn that the live wire makes electrical contact with the case.

What happens?

- A** The current flows to earth and the fuse is not affected.
- B** The fuse melts and switches off the circuit.
- C** The metal case becomes live and dangerous.
- D** The metal case becomes very hot.

53. O/N/09/Q30

Which costs the **most** if operated from the same mains supply?

- A** a 5000 W electric cooker used for 1 minute
- B** a 1000 W electric fire used for 10 minutes
- C** a 500 W electric iron used for 1 hour
- D** a 100 W lamp used for 1 day

54. M/J/09/Q29

An electric iron is marked 240 V, 2500 W.

Four fuses are available with values of 5 A, 10 A, 13 A and 30 A.

Which fuse should be used?

- A** 5 A **B** 10 A **C** 13 A **D** 30 A

55. M/J/09/Q28

A student has a chain of 20 lamps. These are wired in series and connected to the mains.

One lamp blows and all the others go out.

The student wants to find the faulty lamp and replace it.

Where should the student begin?

- A** anywhere, because the current was the same in each lamp
- B** at the live end of the chain, because the current was greatest there
- C** at the middle of the chain, because the current was greatest there
- D** at the neutral end of the chain, because the current was least there

56. O/N/08/Q31

An electric heater is rated at 3 kW. Electrical energy costs 20 cents per kWh.

What is the cost of using the heater for five hours?

- A** 12 cents **B** 60 cents **C** 100 cents **D** 300 cents

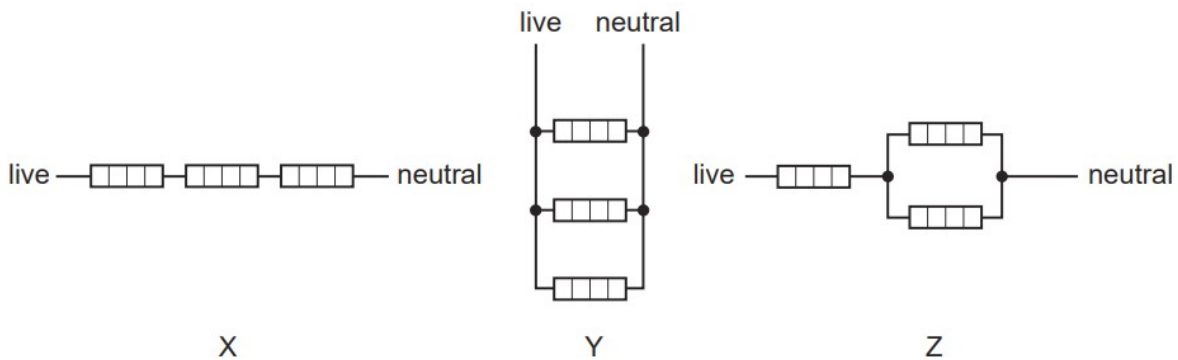
57. O/N/08/Q32

What is the purpose of a circuit breaker in an electric circuit?

- A** to change alternating current into direct current
- B** to keep the current constant
- C** to prevent the current from becoming too large
- D** to reduce the current to a safe value

58. M/J/08/Q30

Three identical heating elements are wired up to the mains supply in the three arrangements shown.



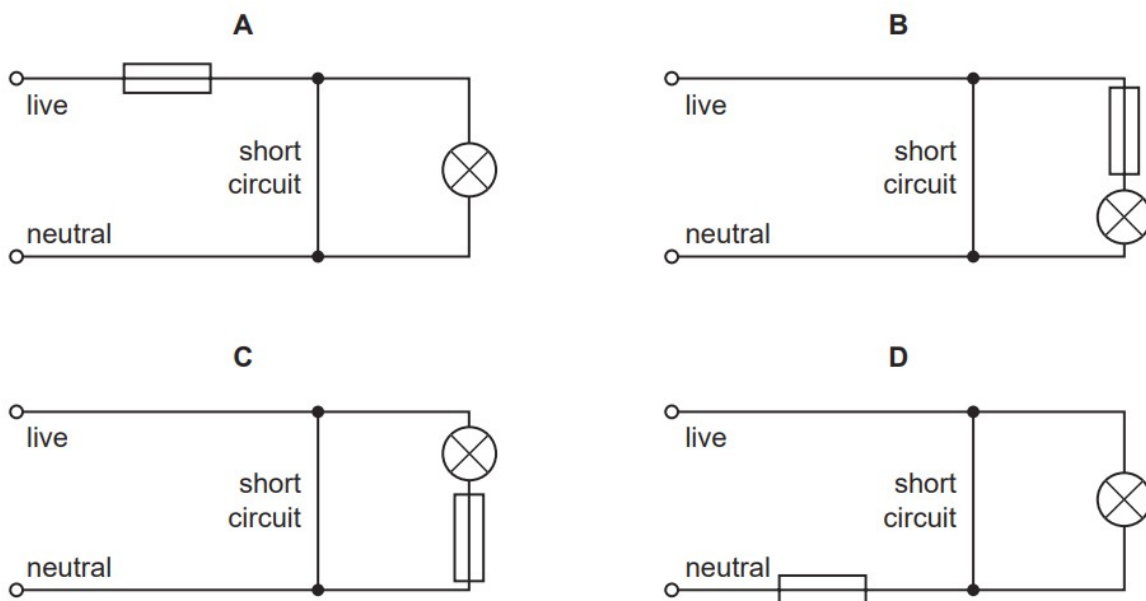
In which arrangement is the current from the supply lowest and in which is it highest?

	lowest current	highest current
A	X	Z
B	X	Y
C	Y	X
D	Y	Z

59. M/J/08/Q32

In each of the circuits below there is a short circuit.

In which circuit does the fuse blow and make the circuit safe to repair?



60. O/N/07/Q31

Energy is represented by the letter E , current by I , power by P , charge by Q , p.d. by V and time by t .

Which pair of equations is correct?

- A** $E = It$ and $P = VI$
- B** $E = VQ/t$ and $P = VI$
- C** $E = VI$ and $P = VI$
- D** $E = VQ$ and $P = VI/t$

61. O/N/06/Q30

A house-owner replaces a failed fuse for the lights of the house. When the lights are switched on, the second fuse also fails. The house-owner then uses a third fuse with a higher rating than the previous two.

Why is this **not** a sensible thing to do?

- A** Fuses only allow the circuit to work if the rating is exactly right.
- B** The third fuse will melt because the rating is too high.
- C** Using a fuse with too high a rating causes electric shocks.
- D** The circuit may work, but the fault is not corrected.

62. O/N/05/Q30

A combined bathroom unit of a heater and a lamp is controlled by one switch. The unit contains a 2 kW heater and a 100 W lamp. In one week, the lamp uses 1 kWh of electrical energy.

How much electrical energy is used by the heater alone?

- A** 2 kWh
- B** 4 kWh
- C** 10 kWh
- D** 20 kWh

63. O/N/05/Q31

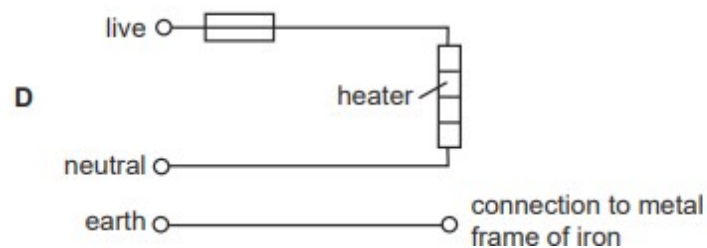
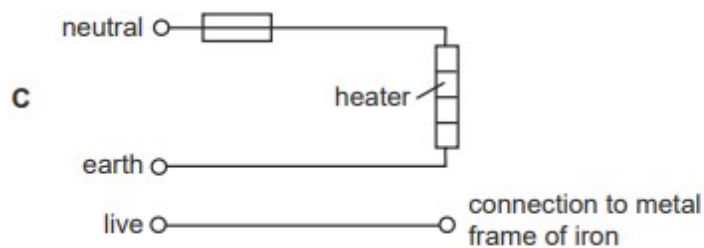
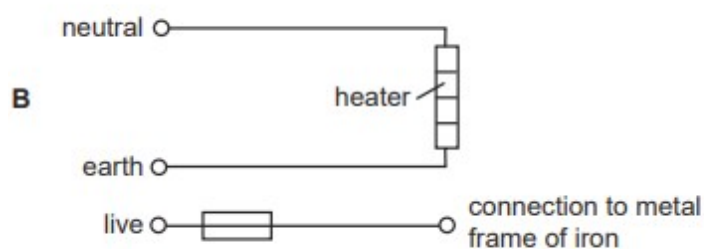
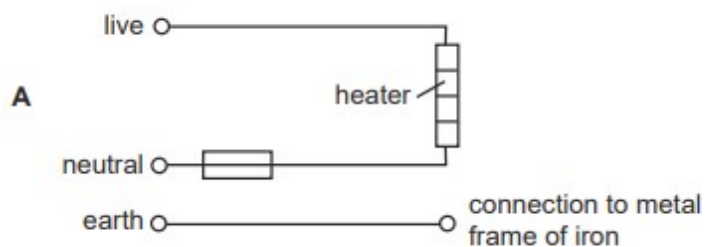
An electric kettle is plugged in and switched on. The fuse in the plug blows immediately.

Which single fault could cause this?

- A** The earth wire is not connected to the kettle.
- B** The live wire and neutral wire connections in the plug are swapped around.
- C** The live wire touches the metal case of the kettle.
- D** The wires connected to the plug are too thin.

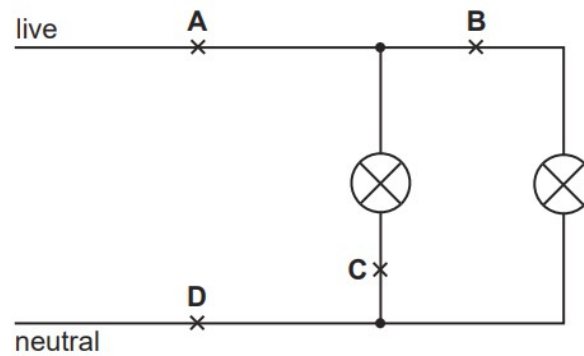
64. O/N/04/Q32

Which circuit shows how the heater in an electric iron and a fuse should be connected to a mains electricity supply?



65. M/J/04/Q30

In order to turn off both lamps, which is the safest switch position?



ANSWERS

- | | |
|-------|-------|
| 1. B | 35. B |
| 2. A | 36. B |
| 3. A | 37. D |
| 4. B | 38. D |
| 5. D | 39. B |
| 6. D | 40. D |
| 7. C | 41. B |
| 8. C | 42. A |
| 9. A | 43. C |
| 10. C | 44. B |
| 11. C | 45. C |
| 12. C | 46. D |
| 13. A | 47. C |
| 14. C | 48. D |
| 15. D | 49. C |
| 16. D | 50. C |
| 17. A | 51. B |
| 18. A | 52. B |
| 19. B | 53. D |
| 20. D | 54. C |
| 21. C | 55. A |
| 22. A | 56. D |
| 23. A | 57. C |
| 24. D | 58. B |
| 25. D | 59. A |
| 26. B | 60. C |
| 27. C | 61. D |
| 28. A | 62. D |
| 29. C | 63. C |
| 30. C | 64. D |
| 31. B | 65. A |
| 32. D | |
| 33. B | |
| 34. B | |

1. O/N/21/22/Q4

A washing machine is working normally with both the water heater and the motor switched on. The washing machine is connected to the mains supply by a cable.

- (a) The current in the live wire in the cable is 13A.

State the size of the current in:

- (i) the neutral wire in the cable

current in neutral wire = [1]

- (ii) the earth wire in the cable.

current in earth wire = [1]

- (b) The insulation on the mains cable is now damaged and, as the washing machine vibrates, the live wire touches the metal casing.

- (i) Explain how the earth wire and the fuse together prevent any more damage.

.....
.....
.....
.....
..... [3]

- (ii) Explain why it is the live wire into which the fuse is connected.

.....
.....
..... [2]

[Total: 7]

2. O/N/21/22/Q9

A filament lamp is connected to a 240 V alternating current (a.c.) mains supply.

- (a) Describe how the output of an a.c. supply differs from the output of a direct current (d.c.) supply.

Sketch **two** voltage–time graphs in the blank space to help your explanation.

.....
..... [3]

- (b) The lamp is rated at 60W and is designed to be used with a 240V supply.

- (i) Calculate the current in the lamp.

current = [2]

- (ii) Calculate the resistance of the lamp.

resistance = [2]

- (iii) A room is lit by five of these filament lamps connected in parallel.

State **two** advantages of connecting the lamps in parallel rather than in series.

1.
.....
2.
.....

[2]

- (iv) The five lamps in (b)(iii) are lit for an average time of 5.5 hours a day for a year. Electricity costs \$0.15/kWh.

Calculate the cost of using these lamps for a year.

cost = [2]

- (c) A student takes the lamp in (b) to school and connects it in a circuit using a 1.5V cell, an ammeter and a voltmeter. The circuit is used to determine the resistance of the lamp.

- (i) In the blank space, draw the circuit diagram of the circuit used to determine the resistance of the lamp.

[2]

- (ii) The value of the resistance of the filament lamp in this circuit differs greatly from the value calculated in (b)(ii).

State how the resistance value in this circuit differs and explain why it differs.

.....
.....
..... [2]

[Total: 15]

3. O/N/20/22/Q5

The cable of a washing machine contains three separate wires. There is a fuse in one of the wires.

- (a) Explain how the earth wire and the fuse work together to make the washing machine safer.

.....

.....

.....

.....

..... [3]

- (b) (i) State the name of the wire in which the fuse is connected.

..... [1]

- (ii) Explain why the fuse is connected into this wire.

.....

..... [1]

- (c) The cable of a hair-dryer contains only **two** wires.

- (i) State the name of each of these wires.

1.

2.

[1]

- (ii) Suggest why the hair-dryer does not need an earth wire.

.....

..... [1]

[Total: 7]

4. M/J/20/22/Q6

The clothes iron shown in Fig. 6.1 is connected to the electrical mains.

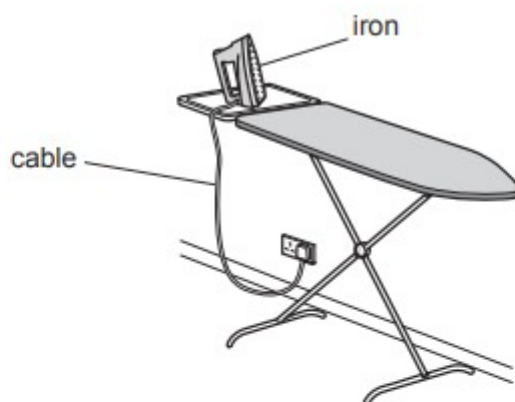


Fig. 6.1

- (a) The boxes in the left column below contain some electrical hazards. The boxes in the right column contain methods of protection from these hazards.

For each hazard, draw **one** line to the appropriate method of protection.

electrical hazard	method of protection
worn insulation on the cable to the iron	earth wire and fuse in plug correctly connected to iron
loose live wire in the iron touches its metal case	circuit breaker correctly connected in circuit
cable becomes too hot because current is too high	visual check of cable before connecting to mains

[2]

- (b) The power of the iron is 1200W. The cost of 1 kWh of electrical energy is 20 cents (20c).

- (i) Define the *kilowatt-hour* (kWh).

.....
 [1]

- (ii) The iron is on at full power for 20 minutes.

Calculate the cost of running the iron for this time.

cost = [2]

5. M/J/18/22/Q8

A fuse is one form of protection in an electrical circuit.

- (a) State **two other** forms of protection that are included in household electrical circuits.

These may protect the consumer, the circuit or an electrical appliance.

1.

2.

[2]

- (b)** Fig. 8.1 shows a fusebox connected to part of a lighting circuit in a house.

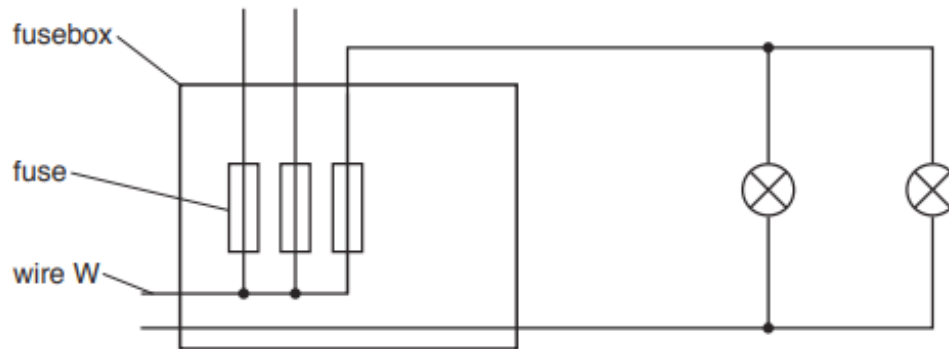


Fig. 8.1

- (i) State how Fig. 8.1 shows that wire W is the live wire.

.....[1]

- (ii) On Fig. 8.1, mark with a letter s, the correct position for a switch that controls both lamps. [1]

- (iii) The rating of the fuse in the lighting circuit is 5A.

Explain what this means.

.....

.....[1]

6. M/J/17/21/Q6

Fig. 6.1 shows a 240 V a.c. mains supply connected to a television and two lamps.

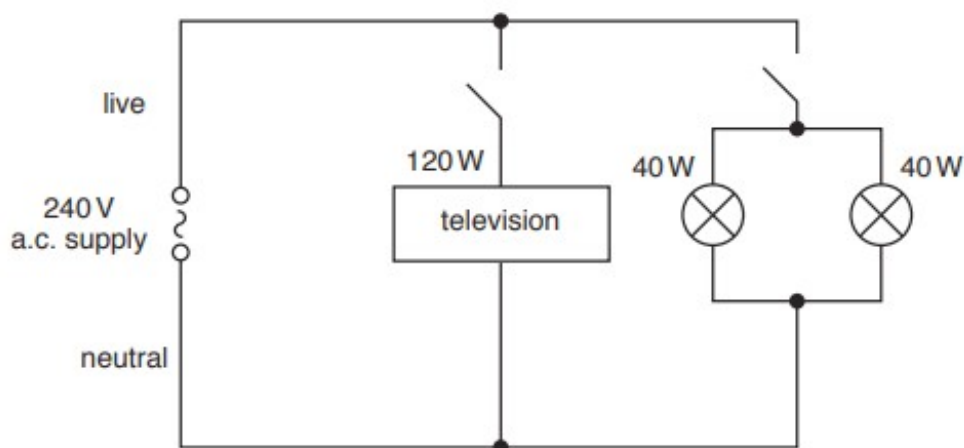


Fig. 6.1

In normal operation, the power supplied to each lamp is 40 W and the power supplied to the television is 120 W.

(a) Calculate, in normal operation,

(i) the total power supplied,

power =[1]

(ii) the total number of kilowatt-hours (kWh) of energy supplied to the circuit in 3.0 hours,

number of kWh =[2]

(iii) the current in each lamp.

current =[2]

(b) Explain why the switches are placed in the live wire and not in the neutral wire.

.....

[2]

7. O/N/16/21/Q7

The a.c. mains electricity system in a house consists of three wires: the live, the neutral and the earth.

- (a) (i)** State the purpose of the live wire.

.....
.....[1]

- (ii)** Explain why the neutral wire is needed.

.....
.....[1]

- (b)** Each circuit in the house includes a fuse for protection.

- (i)** An electrical appliance is switched on and the current in the fuse becomes greater than the rating of the fuse.

State and explain what happens.

.....
.....
.....[2]

- (ii)** State and explain in which of the three wires the fuse is located.

.....
.....
.....
.....[2]

8. O/N/15/22/Q8

A spotlight in a television studio is at full brightness. The lamp inside the spotlight is powered by a 230 V supply and the current in the filament of the lamp is 27 A.

(a) Calculate the power of the lamp.

power = [2]

(b) The spotlight is kept switched on at full brightness.

(i) Calculate the energy transformed by the lamp in 30 minutes.

energy = [1]

(ii) The cost of using one kilowatt-hour (kWh) of electricity is 23 cents.

Calculate the cost of using the spotlight for 30 minutes.

cost = [2]

9. M/J/15/22/Q5

When a balloon is rubbed on hair, the balloon becomes negatively charged. The balloon is shown in Fig. 5.1.



Fig. 5.1

- (a)** Explain how rubbing causes the balloon to become negatively charged.

.....

.....

..... [2]

- (b)** Explain why the hair is pulled towards the balloon.

.....

.....

.....

..... [2]

- (c)** Explain why it is important that the balloon is made from an electrical insulator.

.....

..... [1]

10. M/J/15/22/Q7

An electric hairdryer and an electric heater are connected to the mains supply, as shown in Fig. 7.1.

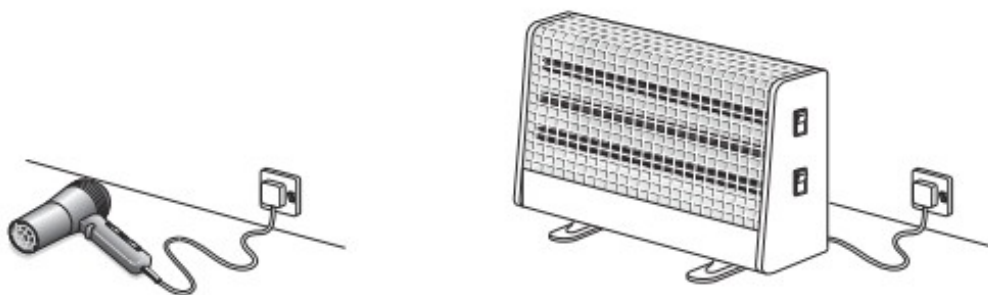


Fig. 7.1

The cable from the heater to the mains supply has a live, a neutral and an earth wire.

- (a)** State the purpose of the neutral wire.

.....
..... [1]

- (b)** The live wire in the electric heater touches the outer metal case.

Explain how the earth and the fuse together protect the user from electric shock.

.....
.....
.....
..... [2]

- (c)** The hairdryer does not have an earth wire. Explain why this hairdryer is still safe to use.

.....
..... [1]

- (d)** In some modern homes, circuit breakers are used instead of fuses.

Suggest one advantage of using a circuit breaker rather than a fuse.

.....
..... [1]

11. O/N/13/21/Q7

A 9.6 kW electric shower is powered by a 240 V mains supply.

- (a) (i) Calculate the electric current in the heating element of the shower.

current = [3]

- (ii) Suggest an appropriate rating for the circuit breaker in the circuit for the shower.

..... [1]

- (b) A family uses the shower for a total time of 25 minutes every day. The cost of 1 kWh of electrical energy is 21 cents.

Calculate the daily cost of using the shower.

cost = [2]

12. M/J/12/21/Q9

Fig. 9.1a shows a room heater. Fig. 9.1b is a diagram of the electric circuit of the heater.

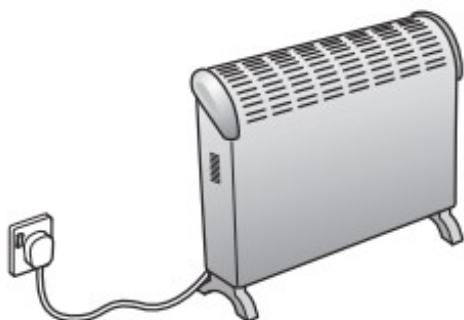


Fig. 9.1a

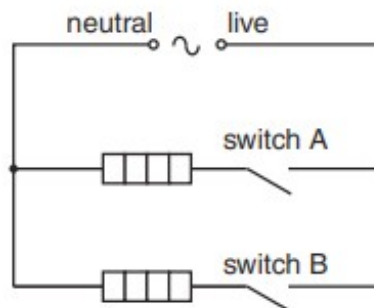


Fig. 9.1b

The fuse has not been drawn on the circuit diagram in Fig. 9.1b.

- (a) (i) On Fig. 9.1b, draw the symbol for a fuse in the correct position. [2]

- (ii) State the part of the room heater to which the earth wire is connected.

.....[1]

- (iii) The earth wire reduces the chance of an electric shock if a fault develops in the room heater.

1. State one fault that causes an electric shock when a person uses the room heater without an earth connection.

.....
[1]

2. Explain how using an earth connection prevents an electric shock.

.....

[2]

- (b) (i) This type of room heater is very efficient. Explain what this means.

.....
[1]

- (ii) The room heater is a convector heater. Describe and explain how thermal energy (heat) passes around a room by convection.

.....

[3]

- (c) Fig. 9.2 shows the power output of the room heater when each switch is closed.

	power / W
switch A only closed	600
switch B only closed	
both switches closed	2100

Fig. 9.2

- (i) Determine the power output of the room heater when only switch B is closed.

power output = [1]

- (ii) The room heater is used with both switches closed for 2.5 hours.

Calculate the energy output of the room heater

1. in kilowatt-hours,

energy = kWh [2]

2. in joules.

energy = J [2]

13. O/N/11/21/Q10

Fig. 10.1 is the circuit diagram of a fan oven that works from a 230V a.c. mains supply.

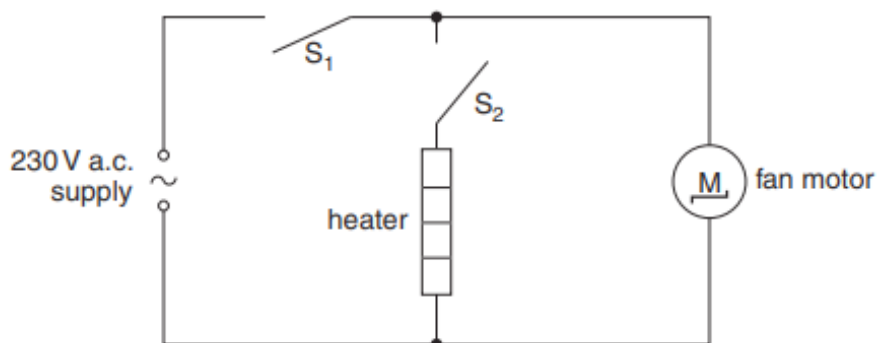


Fig. 10.1

The oven contains a 3500W electric heater, an electric fan operated by a 200W fan motor and two switches S_1 and S_2 .

- (a) (i) When switches S_1 and S_2 are open, both the heater and the fan motor are off, as shown in Fig. 10.2.

S_1	S_2	heater	fan motor
open	open	off	off
		off	on
		on	on

Fig. 10.2

Complete Fig. 10.2 by writing "open" or "closed" in the boxes. [2]

- (ii) Explain how the circuit represented by Fig. 10.1 ensures that the heater cannot be switched on when the fan motor is off.

.....

 [2]

- (iii) Suggest one reason why the circuit is designed so that the heater cannot be switched on when the fan motor is off.

.....
 [1]

- (b) (i) The heater and the fan motor are both switched on. Calculate the current supplied to the oven.

current =[2]

- (ii) 1. Suggest a suitable rating for the fuse to be used with this oven.

.....[1]

2. The fuse is placed in one of the three wires in the mains lead. State the wire in which it is placed.

.....[1]

- (iii) Explain how earthing the metal case of the oven makes it safer.

.....
.....
.....
.....[3]

- (c) The cost of 1 kWh of electrical energy is 35 cents. Calculate the cost of leaving just the fan motor switched on for 12 hours.

cost =[3]

14. M/J/11/22/Q6

The cable from the mains plug to a washing machine contains a live wire, a neutral wire and an earth wire. The earth wire is connected to the metal case of the washing machine.

- (a)** Explain how connecting the earth wire to the metal case makes the washing machine safer.

.....

.....

.....[2]

- (b)** When in use, the average input power to the washing machine is 500W.

Calculate the number of kWh of energy used by the washing machine in 45 minutes of use.

number of kWh =[2]

15. O/N/09/Q6

A microwave oven is rated at 650W and is connected to a 230V mains supply.

- (a) (i) Calculate the current from the supply when the microwave oven is switched on.

current = [2]

- (ii) Suggest a rating of the fuse for use with this oven.

fuse rating = [1]

- (b) The insulation of the mains cable has worn away. The live wire touches the outer metal casing of the microwave oven.

- (i) Explain the hazard that results if the outer metal casing is not earthed.

.....
.....
..... [2]

- (ii) Explain how connecting the earth wire to the outer casing and using a fuse of a suitable rating removes this hazard.

.....
.....
.....
..... [2]

16. M/J/06/Q6

Fig. 6.1 shows a mains extension lead. The six sockets allow several electrical appliances to be connected to the mains supply through one cable.

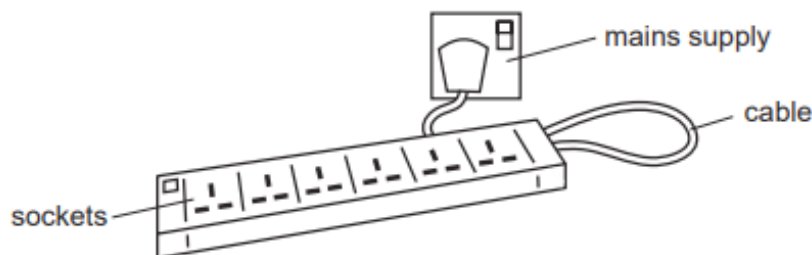


Fig. 6.1

- (a) The cable connects the sockets to the mains supply.

The cable contains three wires: live, neutral and earth. State what is meant by

- (i) *live*,

.....
..... [1]

- (ii) *neutral*,

.....
..... [1]

- (iii) *earth*.

.....
..... [1]

- (b) Six powerful lamps are plugged into the sockets and switched on, one by one.

- (i) State what happens in the cable as the lamps are switched on, one by one.

.....
..... [1]

- (ii) Describe why it can be dangerous when a fuse of the wrong value is used in the plug.

.....
.....
..... [1]

- (c) Explain why your hands should be dry when you put a plug into a socket.

.....
..... [1]

17. M/J/05/Q6

The table gives information about two household appliances.

appliance	mains supply voltage / V	current through appliance / A	power / W	power / kW	time used per day / h	energy used per day / kWh
television	240	1.20	288	0.288	2.50	0.720
water heater	240	12.6			0.50	

(a) Write the missing values in the empty spaces in the table. [3]

(b) Why is more power needed for the water heater than for the television?

.....

 [1]

(c) The water heater is connected to the mains supply. Explain why using a 3 A fuse would **not** be suitable.

.....

 [2]

ANSWERS

1.

Question	Answer	Marks
4(a)(i)	13 A	B1
4(a)(ii)	0 (A)	B1
4(b)(i)	resistance of earth wire (and casing) small or disconnects circuit or no electric shock (possible) or stops current	B1
	current in (live and) earth wire (briefly)	B1
	fuse (in live wire) blows / melts	B1
4(b)(ii)	voltage (source) disconnected (by fuse's blowing) or live voltage immediately next to (switch and blown) fuse or it / live wire at high / maximum voltage / potential	B1
	no part of the washing machine / casing is live or no current to <u>casing</u>	B1

2.

Question	Answer	Marks
9(a)	sinusoidal voltage-time graph labelled a.c. drawn	B1
	voltage-time labelled d.c. graph with horizontal, non-zero line	B1
	an a.c. varies with time or an a.c. changes direction / polarity (at regular time intervals) or a d.c. is constant with time or d.c. is never negative	B1
9(b)(i)	$(I =) P / V$ or 60 / 240	C1
	0.25 A	A1
9(b)(ii)	$(R =) V / I$ or 240 / 0.25	C1
	960 Ω	A1
9(b)(iii)	any two from: if one lamp blows the others stay lit can be operated separately each lamp has the correct voltage across it or lamps glow brightly / normally	B2
9(b)(iv)	(number of units =) $5 \times 60 \times 5.5 \times 365 / 1000$ or 600 or 602.25 or 18(.07)	C1
	\$90 or \$90.34	A1
9(c)(i)	cell, lamp and ammeter in series and no other components	B1
	voltmeter in parallel with lamp	B1
9(c)(ii)	(in this circuit) the resistance is (very much) smaller	B1
	(metal) filament is cooler	B1

3.

Question	Answer	Marks
5(a)	if the live / high voltage wire touches the casing / machine or if the casing / machine becomes live	B1
	large current in earth wire or resistance of earth wire small	B1
	fuse melts or disconnects supply	B1
5(b)(i)	live wire	B1
5(b)(ii)	when the fuse melts the live wire is not connected to any part of the device /is isolated from the mains / supply	B1
5(c)(i)	live and neutral	B1
5(c)(ii)	double insulation or it / casing / hair dryer is plastic	B1

4.

Question	Answer	Marks
6(a)	worn insulation → visual check loose wire in live → earth wire cable too hot → circuit breaker ALL three correct B2 one or two correct B1	B2
6(b)(i)	energy used by (a device of power) 1 kW for 1 hour	B1
6(b)(ii)	(number of kW h =) $1.2 \times 20 / 60$ or 0.4 seen	C1
	8 c	A1

5.

Question	Answer	Marks
8(a)	any 2 from - earthing - contact breaker / circuit breaker - double insulation	B2
8(b)(i)	fuses are in live wire or fuse is connected to W / it	B1
8(b)(ii)	switch drawn between the letter W on the diagram and the left-hand lamp	B1
8(b)(iii)	current is larger than 5 A fuse blows / melts / breaks (circuit) or can only pass currents up to 5 A	B1

6.

Question	Answer	Marks
6(a)(i)	200 W	B1
6(a)(ii)	power $\times$ time or any numerical value of power $\times$ 3	C1
	0.6(0)	A1
6(a)(iii)	(I =) P / V algebraic or numerical	C1
	0.17 A	A1
6(b)	(if in neutral) rest of circuit / lamps / television not live / still high voltage / still 240 V or live has high voltage / 240 V / can cause shock	C1
	(if in neutral) <u>when</u> (switched) <u>off</u> circuit / lamps / television not live / still high voltage / still 240 V	A1

- 7.
- (a) (i) supplies the (mains) e.m.f. / voltage B1
- (ii) to complete the circuit / is at 0 V B1
- (b) (i) the circuit / supply is cut / broken **or** current stops B1
fuse melts / blows / burns B1
- (ii) live wire B1
when it cuts the circuit / melts no part of the appliance is live / no shock B1
- 8.
- (a) $(P =)VI$ **or** 230×27 C1
6200 / 6210 W **or** 6.21 / 6.2 kW A1
- (b) (i) 1.1 / 1.12 / 1.1178×10^7 J B1
- (ii) $6.21 \times 0.5 \times 23$ **or** $6.21 \times 30 \times 23$ **or** $3.1 / 3.105 \times 23$ C1
71 / 71.3 / 71.4 / 71.415 c **or** \$ 0.71 / 0.714 / 0.71415 A1
- 9.
- (a) negative charge moves from hair / person / head to balloon C1
electrons move from hair / person / head to the balloon A1
- (b) hair is positive (at end) B1
opposite charges attract B1
or positive and negative attract
- (c) **charges / electrons** B1
- don't flow away
 - aren't conducted (to earth / person)
- stay on balloon / on insulator

- 10.
- (a) to provide a complete circuit (with live) B1
 or to pass current back to mains
 or provide a return path for the current
- (b) current / charge / electrons flow to earth / earth wire / ground (when live touches case) B1
 fuse melts / blows **and** disconnects circuit / cuts live / stops current B1
- (c) doubly insulated B1
 or **case / body** made of plastic / insulator / not made of metal
 or user cannot touch metal
- (d) (circuit breaker) B1
- turns off / acts fast(er)
 - can be reset
 - easy to see it has tripped / switched
 - can detect small difference between live and neutral currents / small (leakage) current to earth
- 11.
- (a) (i) $(I =)P/V$ or 9.6/240 or 9600 C1
 9600/240 or 0.040 C1
 40 A A1
- (ii) any whole number from 41 to 99 (incl.) **with unit** (A) B1
 (e.c.f. from 0.040 A: 1,2,3 A)
- (b) $9.6 \times 25 \times 21$ or $9.6 \times 25/60$ or $9.6 \times 25/60 \times 21$ or 5040 c or \$50.40 etc. C1
 84 c or \$0.84 or €0.84 or £0.84 or Rs0.84 etc. (85.7/86c from 0.42h) A1

12.

- (a) (i) fuse symbol correct B1
in live wire before junction of two elements B1
- (ii) the (metal) case/outside B1
- (iii) 1. live wire touches case; live touches person B1
2. current goes to earth; current does not go through the person B1
fuse blows B1
- (b) (i) most of the energy output is useful/heat; little energy is wasted; B1
- (ii) hot air rises (**not** heat rises) B1
density of hot air is lower B1
convection current mentioned OR hot air rises and cold air falls B1
- (c) (i) 1500 W B1
- (ii) 1. conversion to kW seen on any power; 2.1 (kW) seen C1
5.25; 5.2; 5.3 (kWh) A1
2. $E = P \times t$ in any form, algebraic or using any power or time e.g. 600×2.5 ,
 600×150 C1
 1.89×10^7 (J) OR $3.6 \times 10^6 \times$ (c)(ii)1. A1

13.

- (a) (i)
- | | | | |
|--------|--------|--|--|
| | | | |
| | | | |
| closed | open | | |
| closed | closed | | |
- B1
- B1
- (ii) S_1 closed $\rightarrow$ motor on B1
 S_1 open $\rightarrow$ heater off B1
- (iii) the heater would overheat/burn/melt **or**
more efficient cooking/circulation described B1

(b) (i)	$(I =) P/V$ or 3700/230 or 3500/230 or 200/230 or 15(.217) 16(.08695)A or 16.1 A	C1 A1
(ii)	integral value: 17 $\rightarrow$ 40 A or up to candidate's (b)(i) + 24 live	B1 B1
(iii)	if case becomes live or if live wire touches case fuse blows live/supply disconnected/case safe to touch/prevents shock/ prevented electrocution	B1 B1 B1
(c)	0.20/3.5/3.7 (kW) or 200/3500/3700 $\times$ 12 $\times$ 35 0.20/3.5/3.7 $\times$ 12 $\times$ 35 or 1470 c or 1554 c or 84000 c 84 c or \$0.84 (allow €/£/R etc.)	C1 C1 A1
14.		
(a)	no shock // no electrocution (if) case becomes live // live touches case	B1 B1
(b)	correct conversion to kW, 0.5 seen // conversion to hours // 0.75 // $\frac{45}{60}$ // $(E =) P \times t$ 0.375 // 0.38 // 0.37 (kW h)	C1 A1
15.		
(a) (i)	$(I =) P/V$ or $P = VI$ or 650/230 2.8 or 2.83 A	C1 A1
(ii)	3, 4, 5, 6 or 7 A only	B1
(b) (i)	casing becomes live/at high voltage current through user/user electrocuted/user shocked	B1 B1
(ii)	fuse blows/melts/breaks fuse in live wire/(microwave) disconnected/circuit broken/no current	B1 B1

16.

- (a) (i) high voltage/where voltage (**not** current) arrives/dangerous (wire) B1
- (ii) zero voltage/safe wire B1
- (iii) zero voltage / connected to ground B1
- (b) (i) (wire) heats up/current increases/electrons move faster C1
- (ii) (wire) melts/causes fire (**not** blows/melts fuse) A1
- (c) avoids electrocution/current through person/water is a conductor B1

17.

- (a) 3024 B1
- 3.024 (or 1/1000 of previous answer) B1
- 1.512 (or 1/2 of previous answer) B1
- (b) smaller resistance **accept** more current B1
- (c) heater uses more than 3A **accept** current 12.6A B1
- causes fuse to melt/blow/burn/break B1